

Microservices with Spring Boot, Spring Cloud

Assessment

1. Explain why REST-based AI APIs are suitable for Java backend integration.

REST-based AI APIs use standard HTTP methods and are easy to connect with Java applications. Java frameworks like Spring Boot provide built-in support for REST communication, making AI integration simple and efficient.

2. Explain how JSON parsing fits into AI response handling.

AI APIs usually return responses in JSON format. JSON parsing helps Java applications extract useful information from the response and display or process it correctly.

3. Explain how environment variables protect API keys.

Environment variables store API keys outside the source code. This improves security by preventing sensitive information from being exposed in code repositories or shared files.

4. Explain how Spring Boot REST endpoints expose AI responses.

Spring Boot REST endpoints receive user requests, send them to the AI API, and return the AI-generated response to the client through HTTP responses.

5. Explain how prompt templates improve chatbot consistency.

Prompt templates provide a predefined structure for user queries. This ensures the chatbot generates more consistent, accurate, and predictable responses.

6. Explain error handling strategies when AI APIs fail.

Error handling includes using try-catch blocks, validating API responses, logging errors, and providing user-friendly messages when the AI service is unavailable or returns an error.
